Hamdan Ashfaq

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EDUCATION

University of California, Irvine & San Diego State University Aug 2026 – Present
Ph.D., Computational Science (Joint Doctoral Program) · Expected 2031 *Irvine & San Diego, CA*

- Doctoral committee chair and mentor: Prof. Mehmet Dogan (SDSU, computational materials physics). Faculty contact at UC Irvine: Prof. Ruqian Wu (condensed matter theory).
- Coursework (Fall 2026): Applied Mathematics for Scientists; Mathematical Modeling; Modern Physics; Quantum Physics. Spring 2027 (planned): Scientific Computing; Advanced Computational PDEs; Condensed Matter Physics; Machine Learning Operations.

Denison University Jan 2023 – May 2026
B.A. Computer Science and B.S. Data Analytics, Philosophy concentration *Granville, OH*

- Cumulative GPA 3.51 / 4.0. Completed both degrees in seven semesters. SAT 1510 / 1600.
- Departmental honor cords in Computer Science and in Data Analytics. Denison Phi Society Inductee, top 7% of the Class of 2026.

Aitchison College Aug 2017 – Jun 2022
Secondary education *Lahore, Pakistan*

RESEARCH EXPERIENCE

San Diego State University, Dogan Group Aug 2026 – Present
Graduate Researcher *San Diego, CA*

- Advisor: Prof. Mehmet Dogan. Developing machine-learned interatomic potentials and running the DFT calculations used to train them.

The Ohio State University, Center for Electron Microscopy & Analysis (CEMAS) May 2024 – Aug 2026
Visiting Fellow, Department of Materials Science and Engineering (May 2026 – Aug 2026) *Columbus, OH*

- Advisor: Prof. Jinwoo Hwang. Simulated 4D-STEM datasets of dry and wet amorphous SiO2 (92K and 120K atom models from Sandia National Laboratories) with Kirkland multislice (TEMSIM) at 200 kV and frozen-phonon averaging, giving labeled diffraction data that experiments cannot provide.
- Wrote the simulation pipeline (SLURM job arrays on the Ohio Supercomputer Center, resumable runs, 4D stack assembly) and computed fluctuation electron microscopy variance curves V(k) to compare with experimental 4D-STEM data.
- Designed the next stage, ML analysis of experimental EMPAD 4D-STEM data of amorphous SiO2 on Si, including substrate removal by low-rank modeling, Bragg peak masking and NMF.

Machine Learning Research Intern (Aug 2025 – Dec 2025) *Columbus, OH*

- Wrote time-series analysis code for an ongoing Hwang lab project that tracks how the microstructure of amorphous semiconductors changes with experimental conditions and thermal processing, using four-dimensional scanning transmission electron microscopy (4D-STEM) data from multiple sample sets.
- Built sequential ML models and autocorrelation analyses to find temporal patterns in structural transformations.
- Ran and tuned the analysis workflows on the Ohio Supercomputer Center across multiple compute nodes.

Research Intern, Intel-funded OSU SURE-5 Program (May 2024 – May 2025) *Columbus, OH*

- Led the self-supervised and unsupervised analysis (K-means clustering, t-SNE) of 65,536 unlabeled 4D-STEM nanodiffraction patterns (multiple GB); the results fed into a Microscopy & Microanalysis 2024 poster and a SURE-5 symposium poster.
- Developed supervised ML models of atomic structure in amorphous semiconductors, streamlined 4D-STEM data processing, and used unsupervised methods to find the most common structural patterns.

Purdue University, Summer Undergraduate Research Fellowship (SURF) May 2025 – Aug 2025
Engineering Research Intern *West Lafayette, IN*

- Advisor: Prof. Nikhilesh Chawla. Built a convolutional neural network with a ResNet-50 backbone and a U-Net++ style decoder that classifies fishbone, lamellar, fine and coarse microstructures in SEM images of Sn-58Bi solder with 90.14% accuracy.
- Cut microstructure analysis from hours of manual expert review to about 30 minutes of automated classification, for solder joints used in electronic packaging.
- Wrote the SEM preprocessing: region-of-interest cropping, CLAHE contrast enhancement, intensity normalization and geometric augmentation.
- Parallelized the preprocessing and TensorFlow data pipeline, raising image throughput 30%.

PUBLICATIONS AND PRESENTATIONS

Journal Articles, Under Review

- Islam, M., Wang, M., Fantin, A., Yoo, G.H., Kim, J.Y., **Ashfaq, H.**, Park, E.S., Maass, R., Wang, Y., Fan, Y., Chung, H.S., Lee, S.C., and Hwang, J. "Linking Topological Medium-Range Order and Density-Wave Coherence to Metallic Glass Ductility." *Acta Materialia*, under review, 2026.
- *Contribution: developed the Swin Transformer V2 classifier of rotational symmetries in individual amorphous nanodiffraction patterns, used to map topological medium-range order across the sample.*

Conference Presentations and Posters

- Patnaik, S., Ganju, E., Torbati Sarraf, H., **Ashfaq, H.**, and Chawla, N. "Harnessing the Power of AI in 3D X-ray Microtomography." Poster, Purdue AI Research Showcase, West Lafayette, Indiana, 2026.
- **Ashfaq, H.**, Wu, J., and Chawla, N. "Imaging and Machine Learning of Defects in Semiconductors." Oral presentation, competitively selected, Purdue SURF Research Symposium, West Lafayette, Indiana, 2025.
- **Ashfaq, H.**, and Hwang, J. "Machine Learning in Materials Science: Analyzing Nanodiffraction Patterns of Amorphous Semiconductors." Poster, OSU SURE-5 Research Symposium, Columbus, Ohio, 2025.
- Islam, M., Kim, J.Y., Yoo, G.H., Im, S., Calderon Ortiz, G., **Ashfaq, H.**, Park, E.S., and Hwang, J. "In-Depth Analysis of Structural Heterogeneity in High Entropy Bulk Metallic Glasses Using 4D-STEM." Poster, competitively selected, Microscopy and Microanalysis 2024 Conference, Cleveland, Ohio.

RESEARCH AND TECHNICAL PROJECTS

Fine-Tuning a Foundation Interatomic Potential for Monolayer MoS2 · GitHub 2026

- Ran 157 PBE calculations in GPAW and fine-tuned MACE-MP-0 on them. Force error on a held-out 600 K MD run dropped from 244 to 19 meV/Å, and fine-tuning beat a from-scratch baseline at every training set size (3 seeds).
- Showed that fine-tuning softens the elastic response and that multi-head replay fine-tuning trades force accuracy for strain accuracy. The model ran 12 ps of stable MD on a 108-atom cell.
- Found that no model trained on pristine cells handles sulfur vacancies; adding 26 vacancy structures to training cut the vacancy force error from 301 to 39 meV/Å.

Finding Order in Disorder: Fold Symmetry in 4D-STEM Patterns · GitHub Aug 2025 – May 2026

Senior capstone, Department of Data Analytics, Denison University *Granville, OH*

- Built *foldsym*, a Python pipeline that detects diffraction speckles and classifies local 2- to 6-fold symmetry in nanodiffraction patterns of CuZr metallic glass, packaged with a CLI, a synthetic data generator, 29 unit tests and CI. Awarded Best Senior Capstone Project, Department of Data Analytics.
- Tested it on synthetic patterns with known labels: the rule-based classifier fails beyond a 9° rotation for 6-fold and 13.5° for 4-fold symmetry, which is why the next step is a self-supervised (SimSiam) model.

Volunteer Food Delivery Platform for a Regional Food Pantry Network · GitHub Jan 2026 – May 2026

- Built and delivered a production React Native and Expo app with Supabase authentication and a real-time backend, free of charge for the partner nonprofit. Owned the login and role-routing architecture used by every user.
- Implemented delivery notifications end to end. When A2P 10DLC carrier registration rules blocked push notifications, switched to an SMS provider that bypasses carrier registration. Also added live GPS location sharing and automated build configuration.

AEP Energy, Client Project (Denison Data Analytics Practicum) · GitHub Aug 2025 – Dec 2025

- Measured how accurate PJM's hourly load forecasts (up to 7 days ahead) are for AEP Energy, by forecast horizon, time of week and weather, in R and Python (MAPE, paired t-tests, chi-square tests, NOAA weather correlations).
- Found that forecasts issued 42 to 48 hours ahead had the lowest error, that PJM overpredicted demand by about 930 MW at the 0 hour horizon, and that errors peak on weekday evenings; recommended the 42 to 48 hour window for customer peak alerts in a final report to the client.

Power-Grid Unit Commitment and Dispatch Optimization 2025

- Formulated and solved a mixed-integer linear program that schedules a thermal and renewable generation fleet over 72 hours on real PJM load and renewable-availability data, with startup costs, minimum stable output, renewable availability limits and a 15% reserve margin.
- Found a base-case optimal cost of $27.96 million. Under high-demand and low-renewable scenarios the fleet fell short of its reserve margin in 53 of 72 hours, with a peak deficit near 233 MW.

Predictive Modeling of Organizational Outcomes on Panel Data 2025

- Used principal component analysis and k-means clustering on a panel of 138 organizations (1998 to 2015) to find group types from twelve behavioral and network indicators.
- Trained a Random Forest on the clusters and organizational covariates to predict above- versus below-median annual outcome counts (AUC 0.655 against a 0.5 baseline), and compared it with logistic regression.

Templated Route-Planning Engine in C++ 2025

- Implemented Dijkstra's algorithm on a templated adjacency-list graph with a hand-built binary-heap priority queue (O(log n) insert, extract-min and decrease-key), for O((n + m) log n) overall.
- Benchmarked a 99-vertex shortest path at 0.0128 s, added parsing that skips malformed records, and fixed an integer-overflow bug by templating identifiers to 64-bit types.

Fair Division Analysis of University Housing Allocation 2025

- Modeled a university housing lottery as an indivisible-goods allocation problem and analyzed Random Serial Dictatorship for strategy-proofness, Pareto efficiency and envy-freeness up to one item (EF1), in light of NP-hardness results for allocations that are both envy-free and Pareto-efficient.
- Designed a rotating-priority mechanism that weights lottery order by cumulative historical dissatisfaction, with a correctness argument and O(nT + n log n) runtime analysis.

TECHNICAL SKILLS

Programming: Python, C/C++, R, SQL, JavaScript; Git, Linux, Jupyter, RStudio.
Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, TIMM; deep learning, self-supervised learning, convolutional neural networks, vision transformers, unsupervised clustering, dimensionality reduction.
Atomistic Modeling: Density functional theory (GPAW, PBE), machine-learned interatomic potentials (MLIPs), equivariant graph neural networks (MACE), ASE, molecular dynamics.
Quantitative Methods: Time-series analysis, statistical and predictive modeling, mixed-integer linear programming, optimization (Gurobi), Monte Carlo and scenario analysis, numerical simulation, multislice electron-scattering simulation.
Data and Computing: NumPy, Pandas, Matplotlib, OpenCV; high-performance computing and supercomputing, CUDA, parallelized data pipelines; PowerBI, Tableau; React, Supabase.

PROFESSIONAL EXPERIENCE

Invicta Counseling May 2023 – Aug 2023
Automation Engineering Intern *Islamabad, Pakistan*

- Built an RStudio and SQL tool that cut manual data processing 40%, automated inventory management with Python and API integration (70% fewer errors), and ran C/C++ system diagnostics that reduced downtime 45%.

Denison University, Information Technology Services May 2023 – Aug 2023
Assistant Manager, Networks and Operations *Granville, OH*

- On a three-person team, upgraded 500 of the campus's 1,700+ Aruba routers to 503H and 515 models and maintained 180+ data closets, keeping 99.9% uptime and halving network downtime.
- Activated 54 uninterruptible power supply units across campus, with a resolution rate above 97%.

Qatar Petrochemical Company (QAPCO) Jun 2021 – Aug 2021
General Engineering Intern *Doha, Qatar*

- Selected as the only secondary-school intern. Worked on multidisciplinary engineering projects to improve production and learned chemical process simulation, safety protocols and distributed control systems.

LEADERSHIP AND SERVICE

Denison University May 2024 – May 2026
Community Advisor *Granville, OH*

- Supported more than 100 residents for 8 to 10 hours a week through community hours, monthly programming and academic, social and personal guidance, and ran biweekly health and safety inspections of more than 500 rooms.

Denison Campus Governance Association Sep 2023 – May 2024
Class Senator, Class of 2026 *Granville, OH*

- Represented more than 680 students on the Finance Committee, which managed a $1.2 million annual budget and approved more than 50 funding requests. Led reforms that increased cultural programming 40%.

State Bank of Pakistan Oct 2021 – Feb 2022
Student Brand Ambassador *Lahore, Pakistan*

- Represented Aitchison College at national financial-literacy events and founded a youth club that engaged more than 500 students in financial-education programming.

AWARDS AND HONORS

- Best Senior Capstone Project, Denison University Department of Data Analytics, 2026.
- Departmental Honor Cords in both Computer Science and Data Analytics, Denison University, 2026.
- Denison Phi Society Inductee, top 7% of the Class of 2026.
- Denison Alumni Award; Department Scholarship, Music (Percussion).
- Purdue SURF Research Fellowship, 2025; Intel-funded OSU SURE-5 Research Fellowship, 2024 and 2025.

COURSEWORK

Computer Science: Algorithm Design and Analysis; Data Structures; Mathematical Foundations of Computer Science; Object-Oriented Programming; Data Systems; Computer Systems; Software Engineering; Computer Organization; Digital Logic Design.
Data Analytics and Statistics: Applied Statistics and Probability; Machine Learning; Deep Learning Theory; Time Series and Analysis; Advanced Predictive Methods; Advanced Prescriptive Methods; Data Visualization; Linear Programming; Practicum in Data Analytics.
Mathematics and Physics: Calculus II and III; Linear Algebra and Differential Equations; Mathematical Modeling Applications; Mathematical Applications in Natural Sciences and Engineering; Logic; Principles of Physics I and II; Atoms and Molecules.

LANGUAGES

Urdu (native) · English (fluent) · Hindi (fluent) · Punjabi (fluent) · Arabic and Persian (read and write) · French (elementary).